

ShopSphere

S3 - Category Management and Navigation

Backend + DB

Platform-Specific Sprint Planning Document

This document intentionally defines only the behavior and delivery scope of the selected platform. It does not mix implementation details from the other platforms.

1. Sprint Definition

Sprint: S3 - Category Management and Navigation

Platform: Backend + DB

Application: ShopSphere

Objective: Implement the backend and database boundary for Category Management and Navigation, including business rules, persistence and API behavior.

This sprint is complete only when the stated use cases, acceptance criteria and tests pass for this platform.

2. Business Use Cases

ID	Use Case	Actor	Precondition	Main Flow	Expected Result
UC-01	Create category	Admin	Authorized role	Enter category metadata, save	Category becomes available according to status.
UC-02	Browse categories	Customer	Category exists and is active	Open category navigation	Active hierarchy is displayed.

3. Detailed Implementation Plan

1. Create hierarchical category model with parent reference, slug, description, image, status and ordering.
2. Implement CRUD, hierarchy validation and cycle prevention.
3. Return only active customer categories.

4. Data / State / Contract Scope

- categories

API / Integration Contract

- Feature APIs for this sprint, versioned under /api/v1.
- Every endpoint must define authentication, authorization, validation, error codes and request correlation.

5. Rules and Expected Behavior

- The platform must rely on the backend as the source of truth for business rules.
- Do not expose stack traces, secrets or internal implementation details to end users.
- Handle loading, empty, validation and error states explicitly.
- Requests that can be retried must be safe from duplicate side effects where the feature requires idempotency.
- Ownership and authorization must be checked server-side for protected data.

6. Acceptance Criteria

- Data model supports the required business relationships and constraints.
- API enforces authentication, authorization, validation and ownership where applicable.
- Business rules are server-authoritative.
- Transactions/idempotency/concurrency controls are used where the feature requires them.
- Controlled error codes and request correlation are implemented.

7. Test Cases

TC	Scenario	Expected Result
TC-S3-01	Duplicate slug	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S3-02	Self-parent	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.

TC	Scenario	Expected Result
TC-S3-03	Circular hierarchy	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S3-04	Inactive parent	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S3-05	Delete category with products	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S3-06	Empty category	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S3-07	API timeout	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.

8. Sprint Exit Checklist

- Implementation completed only for this platform.
- All primary and negative use cases pass.
- API contracts are implemented/consumed as defined.
- Validation and authorization tests pass.
- Loading/empty/error handling is verified where applicable.
- No critical or blocker defect remains.
- Automated tests pass in CI.
- Documentation and contract changes are committed.

9. Out of Scope for This Document

- Implementation behavior of Customer Portal. That behavior belongs in the separate S3 document for that platform.
- Implementation behavior of Admin Portal. That behavior belongs in the separate S3 document for that platform.